VEHICLE RENTAL SYSTEM

Object-Oriented Programming Project Report

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1. Introduction

The Vehicle Rental System is a software application designed to simplify the process of renting vehicles such as cars and bikes. This system helps manage vehicle details, customer information, and rental transactions efficiently.
It replaces manual record-keeping with an automated system, reducing errors and saving time.

2. Objectives

To maintain records of available vehicles
To manage customer details
To handle vehicle rental and return operations
To generate rental bills automatically
To implement real-world OOP concepts

3. Scope of the Project

This system can be used by:
Car rental agencies
Bike rental services
Travel companies
It helps improve efficiency, accuracy, and customer satisfaction.

4. Features of the System

Add, update, and delete vehicle details
Register new customers
Rent vehicles to customers
Return vehicles and update availability
Calculate rent based on duration
Generate bill/receipt

5. OOP Concepts Used

Encapsulation
Data and methods are combined into a single unit (class).
Example: Vehicle class contains both data (name, price) and functions.
Inheritance
Derived classes (Car, Bike) inherit properties from the base class (Vehicle).
Polymorphism
Same function behaves differently depending on context (e.g., rent calculation for car vs bike).
Abstraction
Only essential details are shown to the user, hiding complex logic.

6. System Design

Classes Used
  Vehicle (Base Class)
Car (Derived Class)
  Bike (Derived Class)
Customer
  Rental
Class Description
  Vehicle
Attributes: vehicleID, name, pricePerDay, availability
Methods: displayDetails()
Customer
Attributes: customerID, name, contact
Methods: registerCustomer()
Rental
Attributes: rentalID, customer, vehicle, days
Methods: calculateBill()

7. Algorithm

Start
Display menu options

Add vehicle details
Register customer
Select vehicle for rent
Enter number of days
Calculate total cost
Return vehicle and update availability
Generate bill
End

8. Sample Code (C++)

C++

```
#include <iostream>
using namespace std;

class Vehicle {
public:
   string name;
   int pricePerDay;

   void display() {
      cout << "Vehicle: " << name << endl;
      cout << "Price per day: " << pricePerDay << endl;
   }
};

class Car : public Vehicle {
};

class Customer {
public:
   string name;
};

int main() {
   Car c;
   c.name = "Sedan";
   c.pricePerDay = 2000;

   c.display();
   return 0;
}
```

9. Advantages

Saves time and effort
Reduces manual errors
Easy to manage records
Efficient billing system
Scalable and flexible

10. Limitations

Basic version may not include online booking
Requires system knowledge to operate
Limited security in simple implementations

11. Future Enhancements

Online booking system
Mobile application integration
Payment gateway integration
GPS tracking of vehicles
Database connectivity (MySQL)

12. Conclusion

The Vehicle Rental System is an effective application of Object-Oriented Programming concepts. It demonstrates how real-world problems can be solved efficiently using classes, objects, inheritance, and other OOP principles.